

UNIVERSITY OF ELDORET
COMP 326 CAT 1

1. (a) State what we use for the following in C#. **(2 Marks)**
 - (i) The name of the C# compiler.
 - (ii) The escape sequence for alert (or ringing the computer's bell).
- (b) (i) Which C# data type represents short integers which are positive? **(1 Mark)**
 - (ii) The C# **switch** block applies to which data types? **(1 Mark)**
- (c) Write the output of the following C# code. **(2 Marks)**

```
int x=7, y=0, z=0;
y=++x; z=y++; z-=x;
Console.Write(y); Console.Write(z);
```
2. (a) (i) Write the definition of a C# method named **valid** that receives the mark obtained in an exam (a number with decimal points) and returns either True (if the mark is between 0 and 100) or otherwise False. **(2 Marks)**
 - (ii) Write a single sample statement to call the function. **(1 Mark)**
- (b) Write C# statements to declare an array named **t** for storing 50 float numbers and input the array using a loop. **(3 Marks)**
- (c) Write a C# class named **a** that contains a data member named **b** (for storing characters), a method to input **b**, as well as a parameterized constructor. **(3 Marks)**
3. The details stored for a book in a shop are the unique item code, name, price, as well as the stock, the author, the edition, the publisher and the year of publication. The details stored for an electronic item are unique item code, name, price, the stock, and the discount rate. The operations done on the items include registering a new item, outputting details of an item, selling an item (update stock appropriately), and computing the discount given (for electronics).

Required: Write down a program with appropriate classes using inheritance/polymorphism for the above. **(15 Marks)**

- E N D -